Field Data Collection

& Mobile GIS

Press Space for next page →

Libraries

KoboToolbox

QField

Learning objectives

- 📝 Understand the **landscape of mobile data collection tools**
- 📋 Design and deploy a **digital survey form** in **KoBoToolbox** using best practices
- 📡 Configure **offline data collection** strategies and manage **data synchronization** with desktop systems
- 📍 Integrate **GPS positioning**, **photo capture**, and **multimedia attachments** into field data workflows
- 🗺️ Prepare a **QGIS project for QField** and collect **georeferenced field observations** on a mobile device
- 🔬 Identify the **right tool for your research context**

Workshop agenda

Time	Topic	Description
5 min	Welcome & Introduction	Introductions, goals, participant check-in
5 min	Why Go Digital?	Benefits of mobile data collection over paper-based methods
20 min	KoBoToolbox	Live demo: survey design, skip logic, validation, offline collection
20 min	QField & Mobile GIS	Live demo: QGIS project setup, GPS integration, field data capture
5 min	Choosing the Right Tool	Survey-driven vs. spatial workflows, combining both tools
5 min	Q&A & Wrap-Up	Open questions, shared resources, follow-up support

Why Go Digital?

📄 Paper-Based Challenges

- **Transcription errors** — manual entry introduces mistakes
- **Data loss** — forms get damaged, lost, or incomplete
- **Lag time** — days or weeks before data is usable

📱 Digital Advantages

- **Real-time validation** — catch errors at the point of collection
- **GPS tagging** — automatic location for every record
- **Multimedia capture** — photos, audio, video in the field
- **Offline capability** — no internet? No problem

Today's Toolkit

KoboToolbox

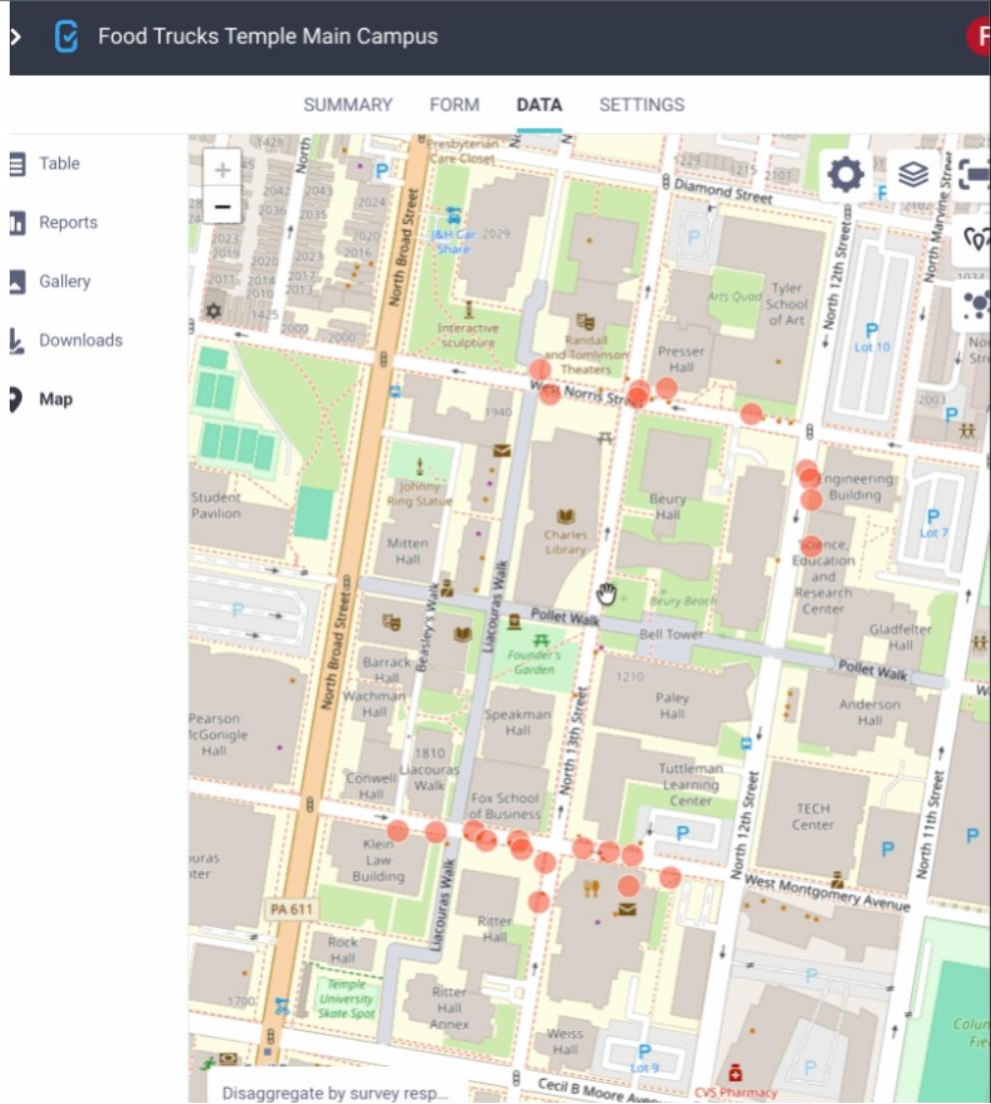
QField for QGIS

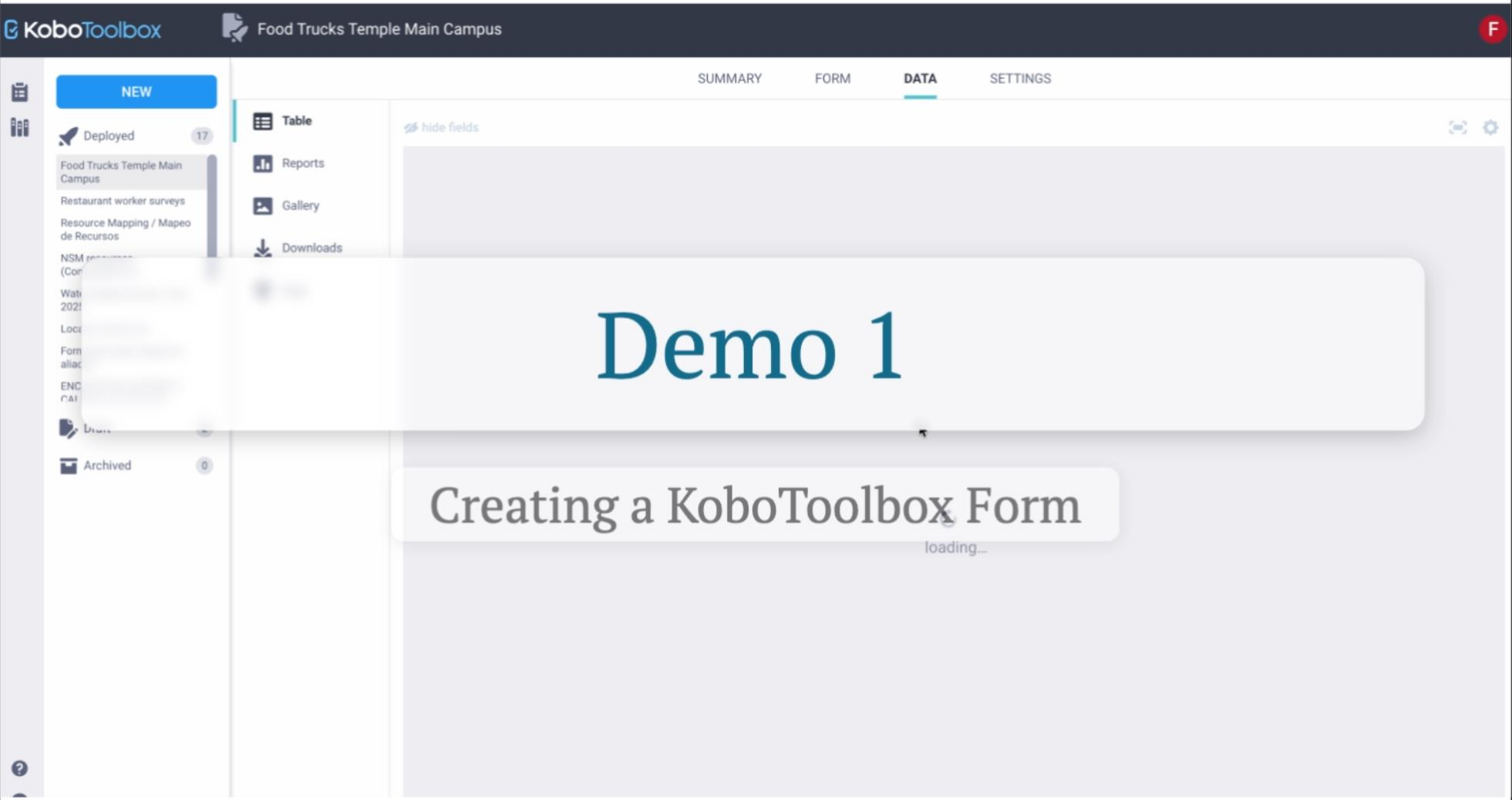
What is KoboToolbox?

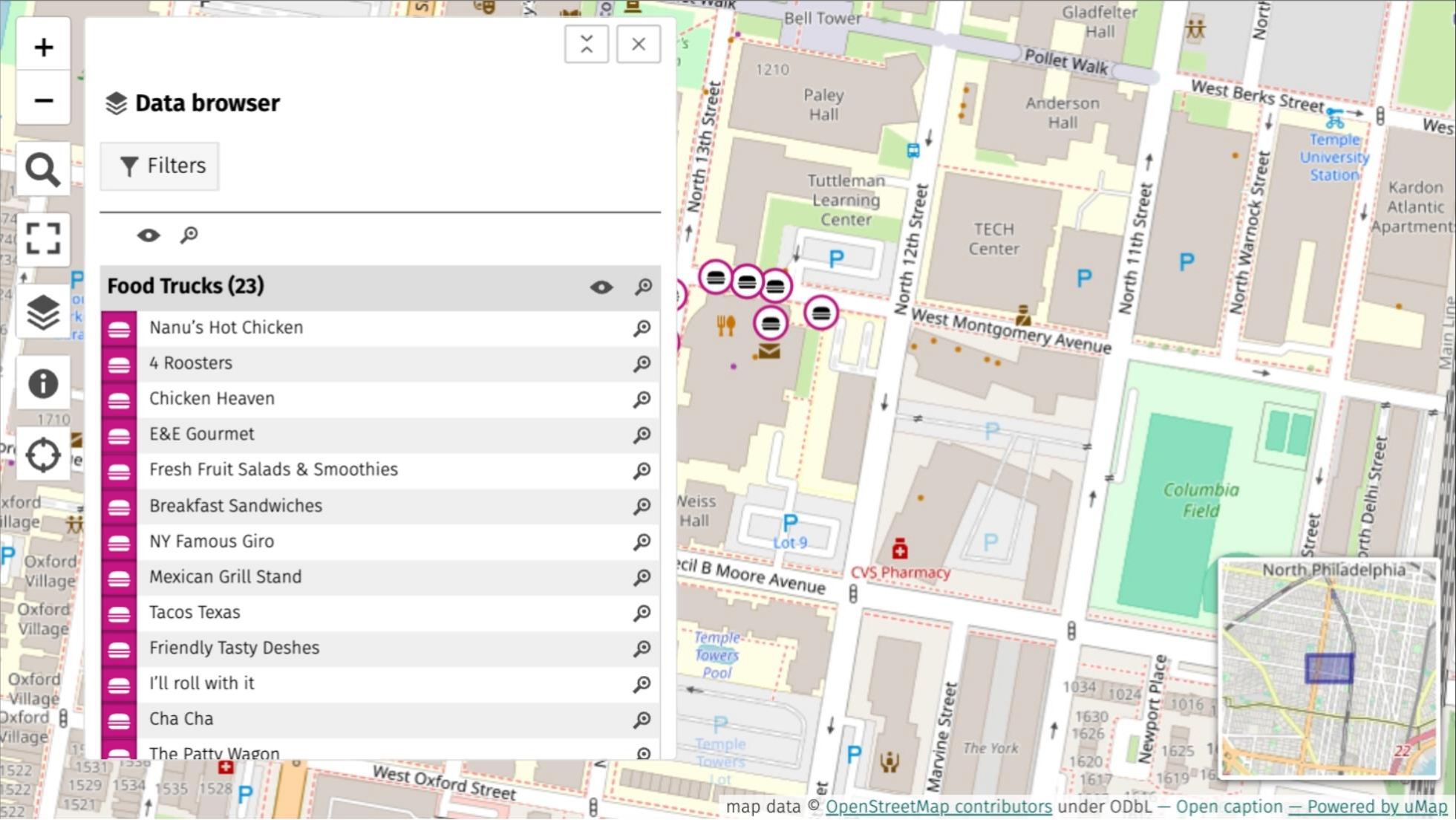
An open source platform for the collection, management, and visualization of data.

KoboToolbox

- Build questionnaires in the web or using XLSForm
- Translate questionnaires into multiple languages
- Create a questions library
- Collect data offline or online in multiple devices (mobile phones, tablets, laptops)
- Visualize data in maps and reports
- Download data in multiple formats (XLS, CSV, KML, ZIP, GeoJSON)
- Share and collaborate on projects




KoboToolbox
Food Trucks Temple Main Campus
F
NEW
Deployed 17
Food Trucks Temple Main Campus
Restaurant worker surveys
Resource Mapping / Mapeo de Recursos
Archived 0
SUMMARY
FORM
DATA
SETTINGS
Table
Reports
Gallery
Downloads
hide fields
loading...
Demo 1
Creating a KoboToolbox Form

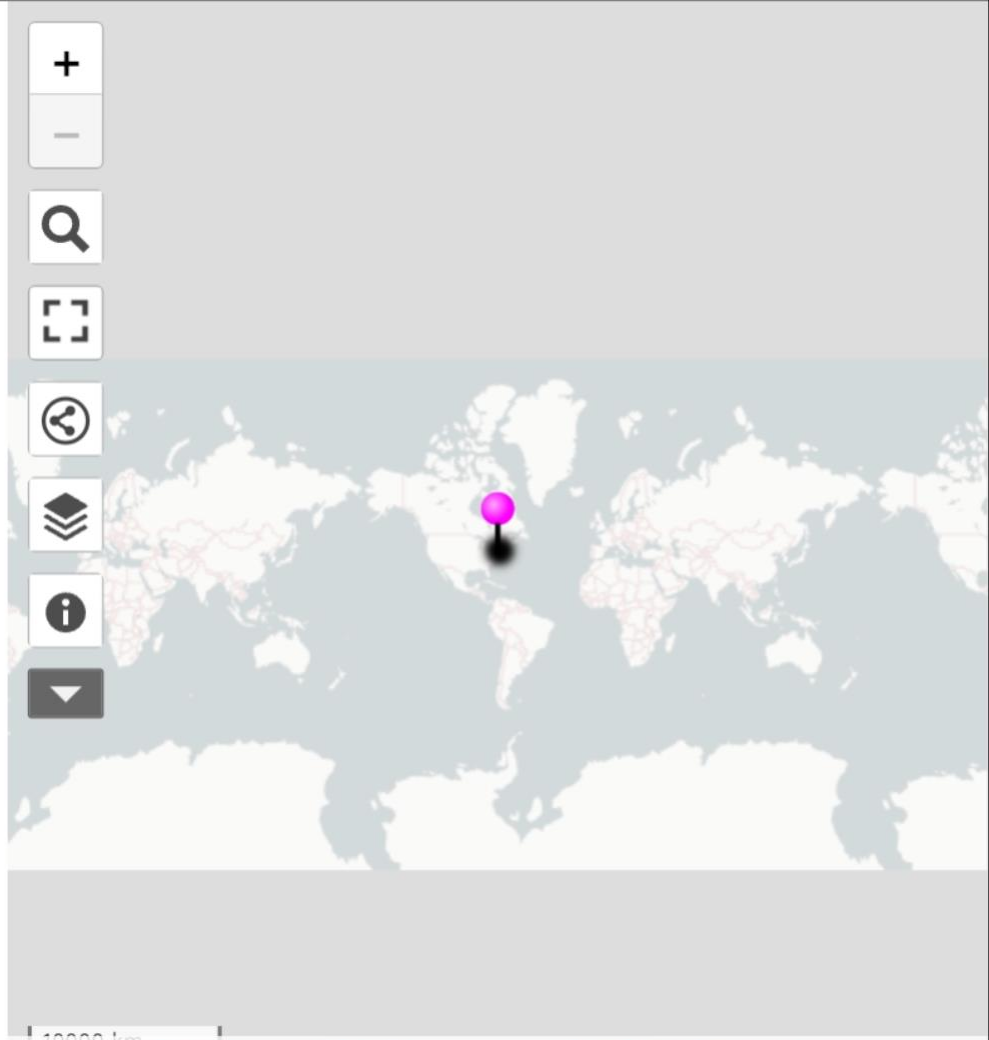

Data browser
Filters
Food Trucks (23)
Nanu's Hot Chicken
4 Roosters
Chicken Heaven
E&E Gourmet
Fresh Fruit Salads & Smoothies
Breakfast Sandwiches
NY Famous Giro
Mexican Grill Stand
Tacos Texas
Friendly Tasty Deshes
I'll roll with it
Cha Cha
The Patty Wagon
Bell Tower
Gladfelter Hall
Pollet Walk
Paley Hall
Anderson Hall
West Berks Street
Temple University Station
Kardon Atlantic Apartments
Tuttleman Learning Center
TECH Center
North 13th Street
North 12th Street
North 11th Street
North Warnock Street
West Montgomery Avenue
Columbia Field
Neiss Hall
Lot 9
CVS Pharmacy
Cecil B Moore Avenue
Temple Towers Pool
Temple Towers Lot
Marvine Street
The York
Newport Place
West Oxford Street
Oxford Village
North Philadelphia
map data © OpenStreetMap contributors under ODbL — Open caption — Powered by uMap

Try it out

Take a look at the sample form for the live map:

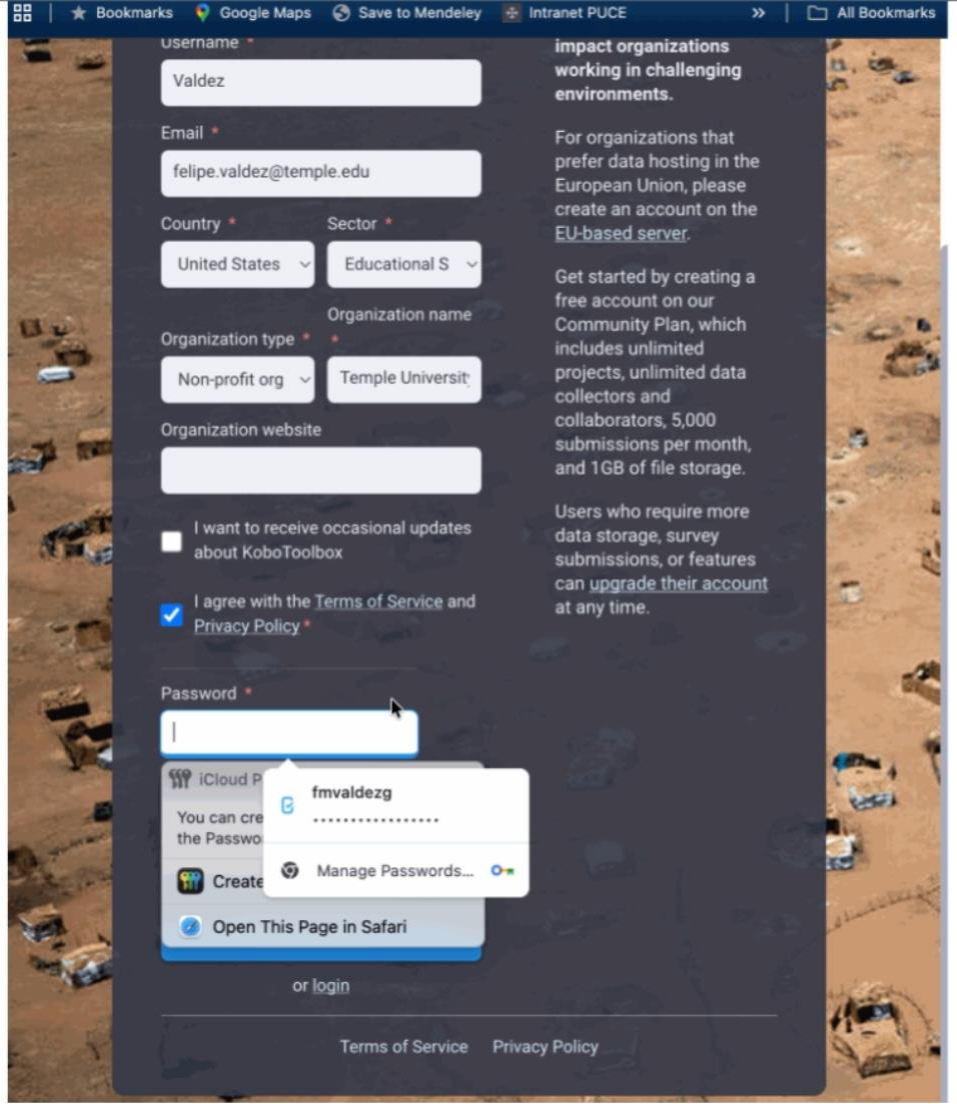
Go to the form

Or scan the QR code:



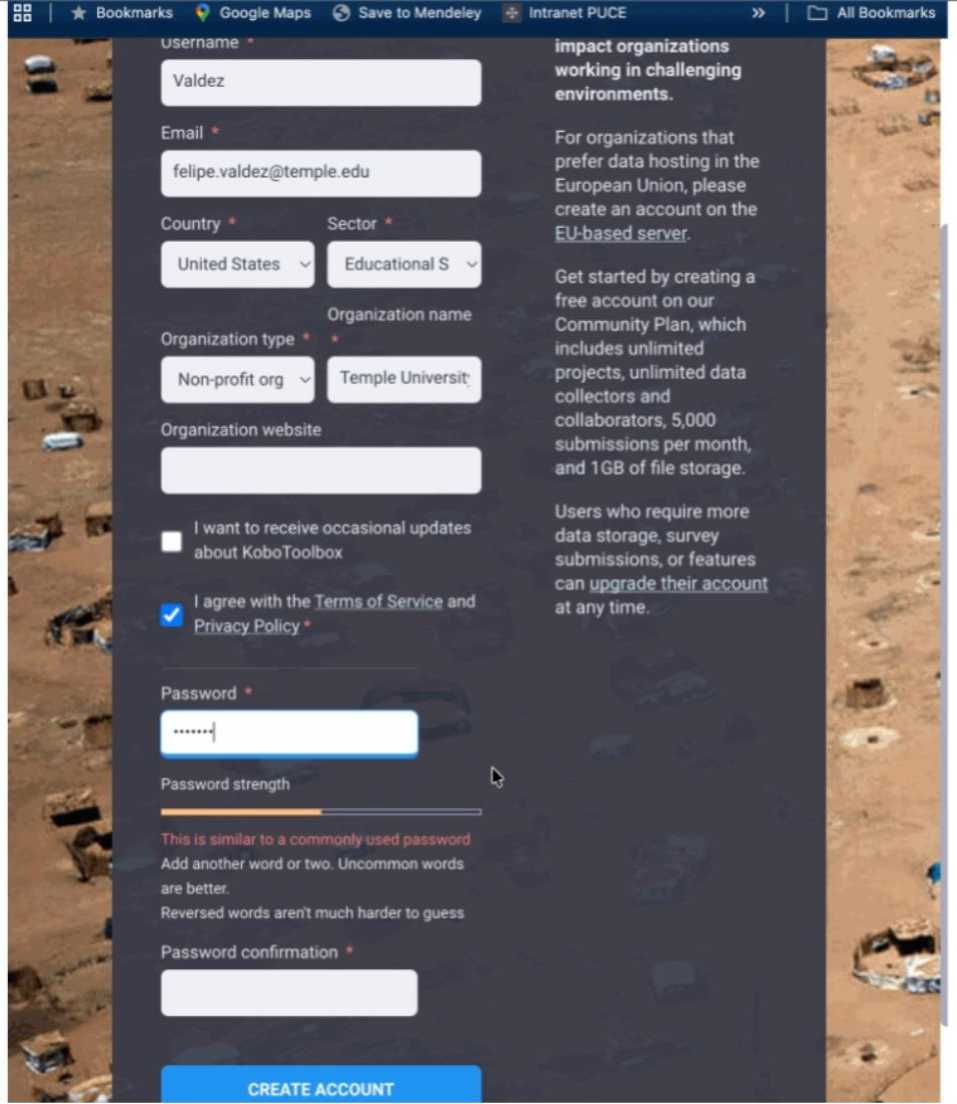
Your turn

1. Go to https://www.kobotoolbox.org/
2. Create an account (use the *Global KoboToolbox Server*)
3. Start a new project
4. Add all types of questions to your form
5. Preview your form
6. Deploy your form when ready
7. Share the link with participants or collect data yourself
8. Analyze your data with reports and maps
9. Download your data



Tips and Tricks

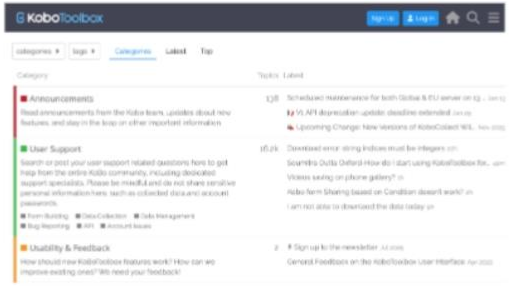
- Always **preview** your form
- Test the form in different devices
- Use alternative languages
- Try out different layouts
- Create the form using XLSForm
- Save the form as an app in your phone/tablet
- Test the form offline (if needed)
- Download the data in different formats
- Configure a REST service to share live updates



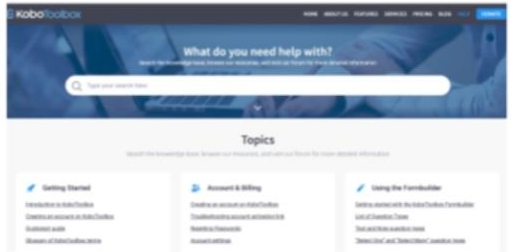
Additional Resources

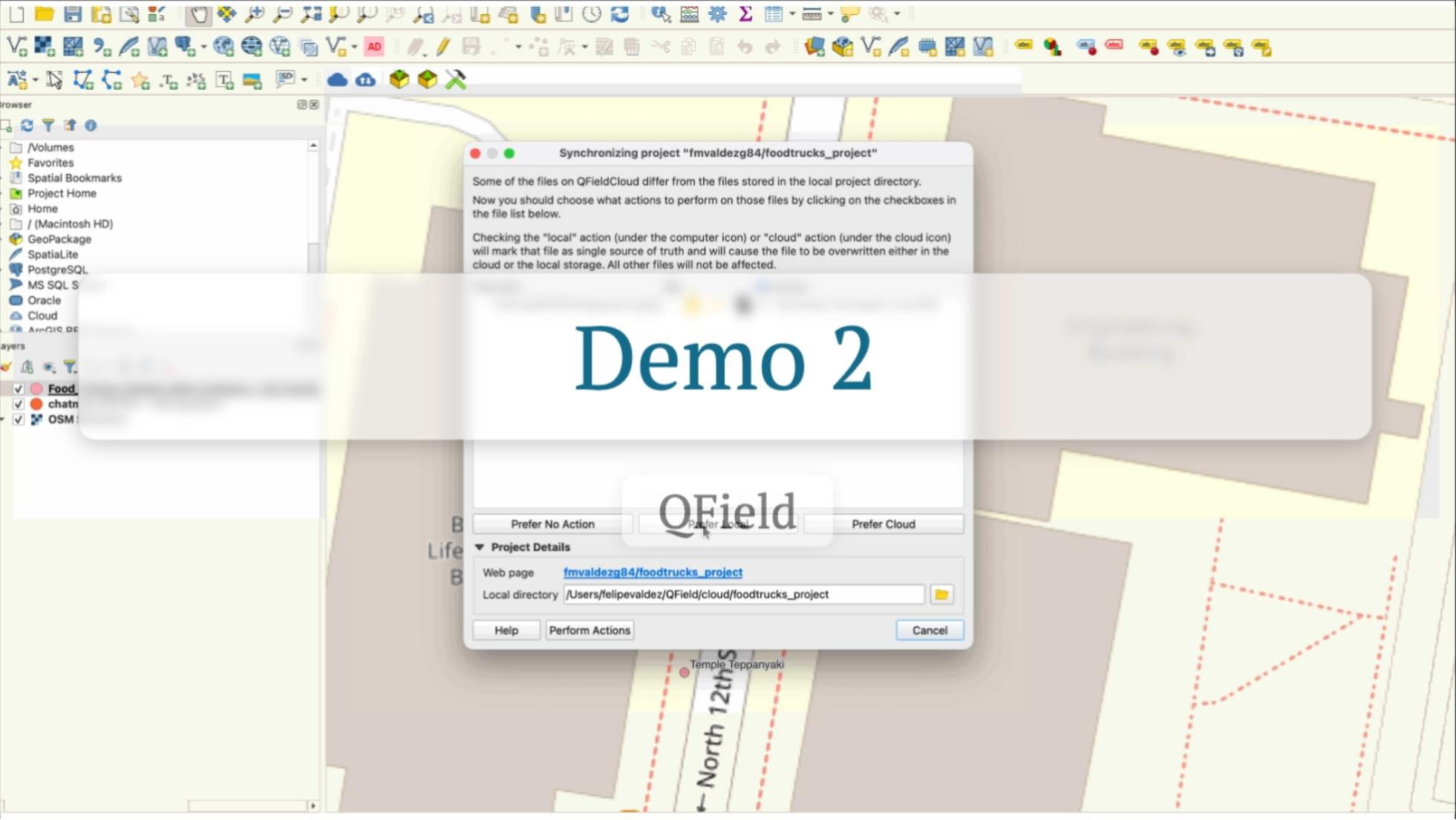
KoboToolbox Academy

KoboToolbox Community



KoboToolbox Help Center

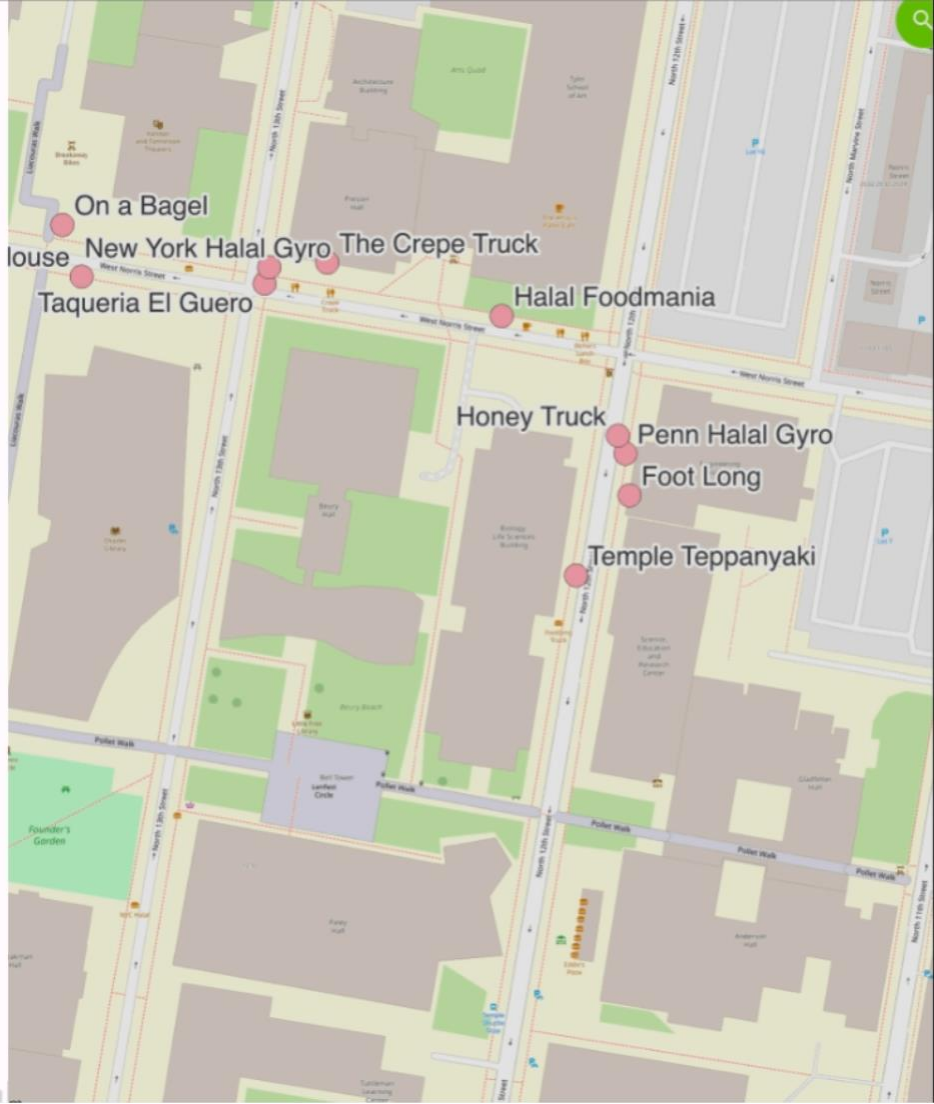



Synchronizing project "fmvaldezg84/foodtrucks_project"
Some of the files on QFieldCloud differ from the files stored in the local project directory.
Now you should choose what actions to perform on those files by clicking on the checkboxes in the file list below.
Checking the "local" action (under the computer icon) or "cloud" action (under the cloud icon) will mark that file as single source of truth and will cause the file to be overwritten either in the cloud or the local storage. All other files will not be affected.
Prefer No Action
Prefer Local
Prefer Cloud
Project Details
Web page
fmvaldezg84/foodtrucks_project
Local directory
/Users/felipevaldez/QField/cloud/foodtrucks_project
Help
Perform Actions
Cancel
Demo 2
QField
Temple Teppanyaki
North 12th S
Favorites
Spatial Bookmarks
Project Home
Home
GeoPackage
SpatiaLite
PostgreSQL
Oracle
Cloud

What is QField?

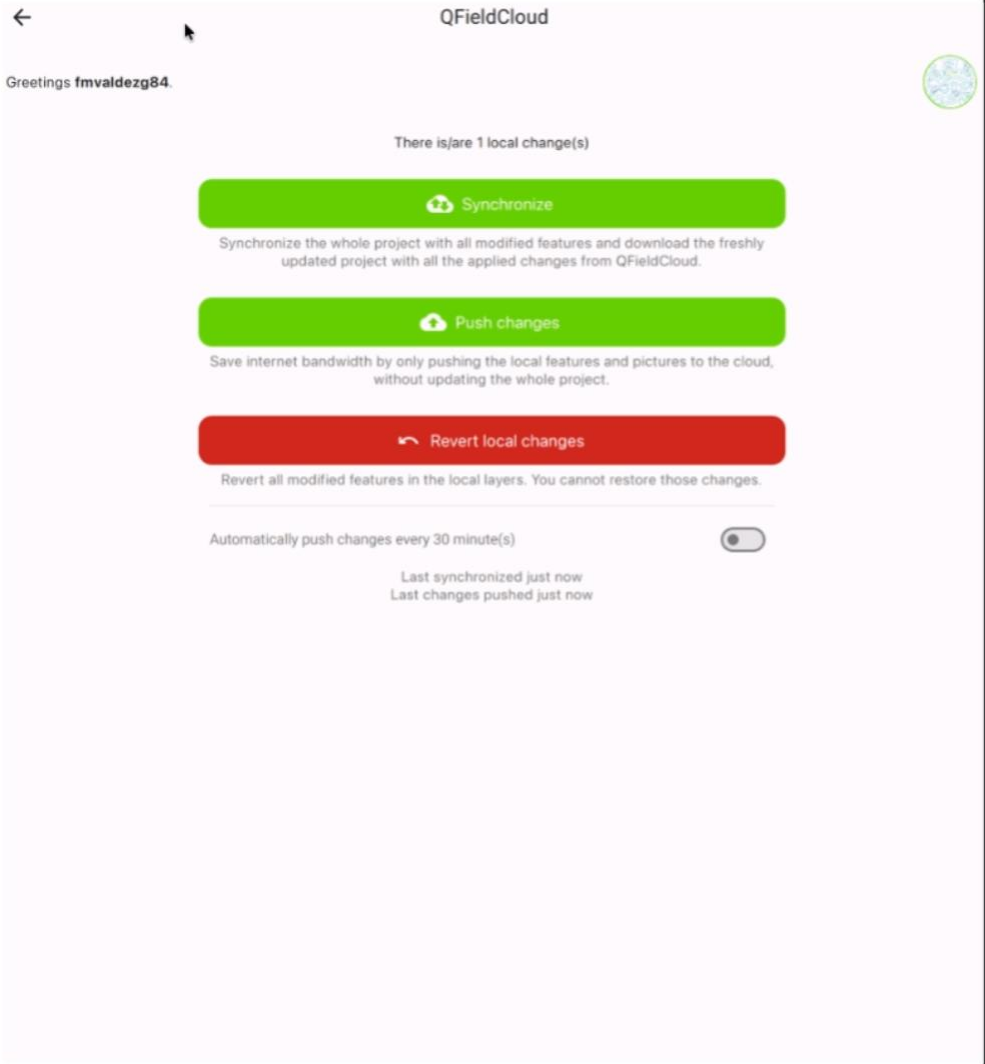
QField is an open-source application that allows painless fieldwork and synchronisation with your local QGIS projects.

- Use it in online and offline
- Link to external databases
- Export from QGIS with the QFieldSync Plugin
- QField is available for Android and iOS devices
- Optimised data collection outside



Your turn

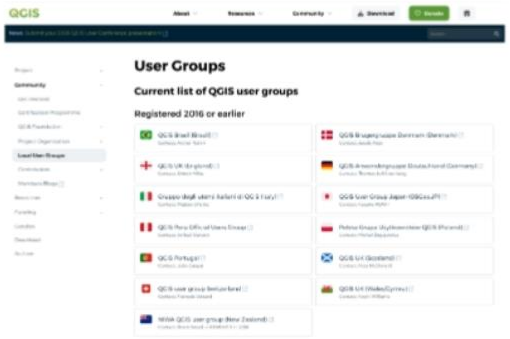
1. Download QGIS
2. Create your project in QGIS
3. Download the QField Plugin
4. Download QField on your mobile device (Phone or Tablet)
5. Export the project to your device
6. Use QField Cloud to synchronize (if needed)
7. Go out and map
8. Update your QGIS project files



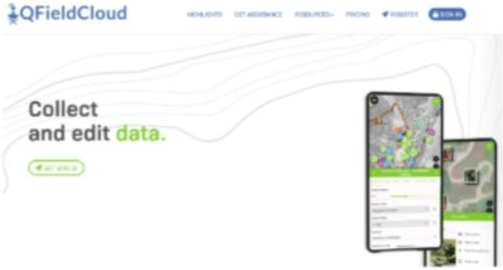
Additional Resources

QField Docs

QGIS user groups



QField Cloud



Temple University Libraries

MAPPING SERIES WORKSHOPS

Introduction to QGIS: Open Source mapping software
Wednesday, January 21, 3pm

Web Mapping for Non-programmers
Wednesday, February 4, 3pm

Mapping Philadelphia in OpenStreetMap
Wednesday, February 11, 12pm

Using OpenStreetMap for Urban Spatial Analysis
Wednesday, February 18, 3pm

Field Data Collection & Mobile GIS
Wednesday, March 11, 3pm

From Historical Maps to Digital Stories
Wednesday, March 25, 3pm

Mapping and Geospatial Data Visualization with Python
Wednesday, April 15, 3pm

Learn more and register at: library.temple.edu/events

Contact us at:
felipe.valdez@temple.edu

Visit our guides:
https://guides.temple.edu/gis-mapping